CYBER SHUJAA

DATA & AI

FINAL EXAM

DATE:	**TIME:** 6:00 PM – 9:00 PM
TIME ALLOWED: 3 Hours	**TOTAL MARKS:** 100

GENERAL INSTRUCTIONS

- Answer ALL questions.
- This exam must NOT be completed in groups or with assistance from others.
- Use this document as your write-up reporting template as you complete each task.
- Rename this file with your full name and Cyber Shujaa ID before submitting.
- Once completed, save and upload to the LMS for marking.
- Before leaving the exam hall, confirm you have uploaded the correct file.
- Compile a detailed report write-up that clearly outlines your approach to each challenge. Write-ups must be authentic and in your own words.
- Include full-screen screenshots for every answer showing: your full desktop with taskbar and timestamp visible, the code you ran, and all outputs.

SECTION A: MULTIPLE CHOICE QUESTIONS (30 Marks)

Note: 6:30PM-7:00PM

SECTION B: PRACTICAL IMPLEMENTATION (70 Marks)

Background / Scenario

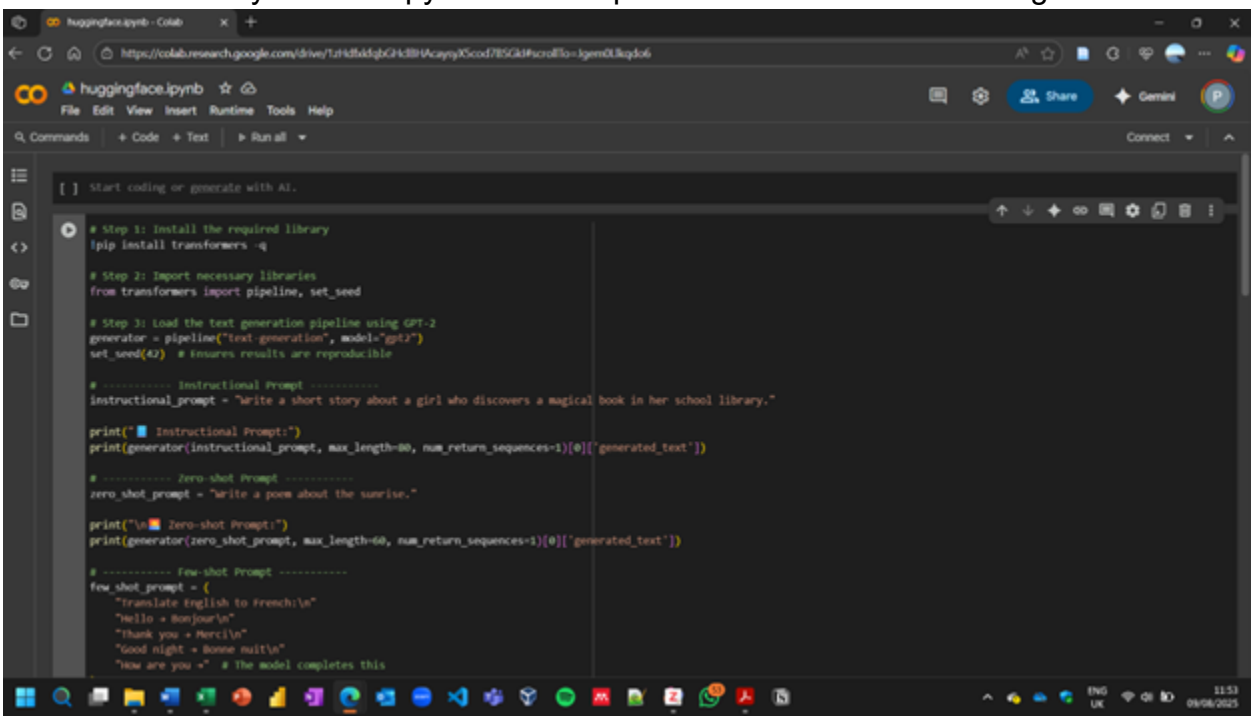
You are a Junior Data & AI Specialist at XYZ Inc., a global technology recruitment and workforce analytics firm. The company collects detailed records on employees across various industries, job roles, and locations worldwide. Your manager has tasked you with building a data pipeline and predictive model to help the company understand salary drivers and predict compensation levels insights that will directly support the firm's salary benchmarking, talent acquisition, and career advisory services.

Dataset: [xyz_inc_salary_prediction.csv]

Documentation Requirements

For ALL practical tasks you must:

- Create a Python Notebook on Google Colab. Use Markdown cells for all notes, explanations, and interpretations at every step.
- Show full-screen screenshots for every answer: full desktop with taskbar and timestamp visible, code executed, and all outputs clearly displayed.
- Document your problem-solving process including challenges faced and how you resolved them.
- Download your final .ipynb file and upload it to the LMS for marking.



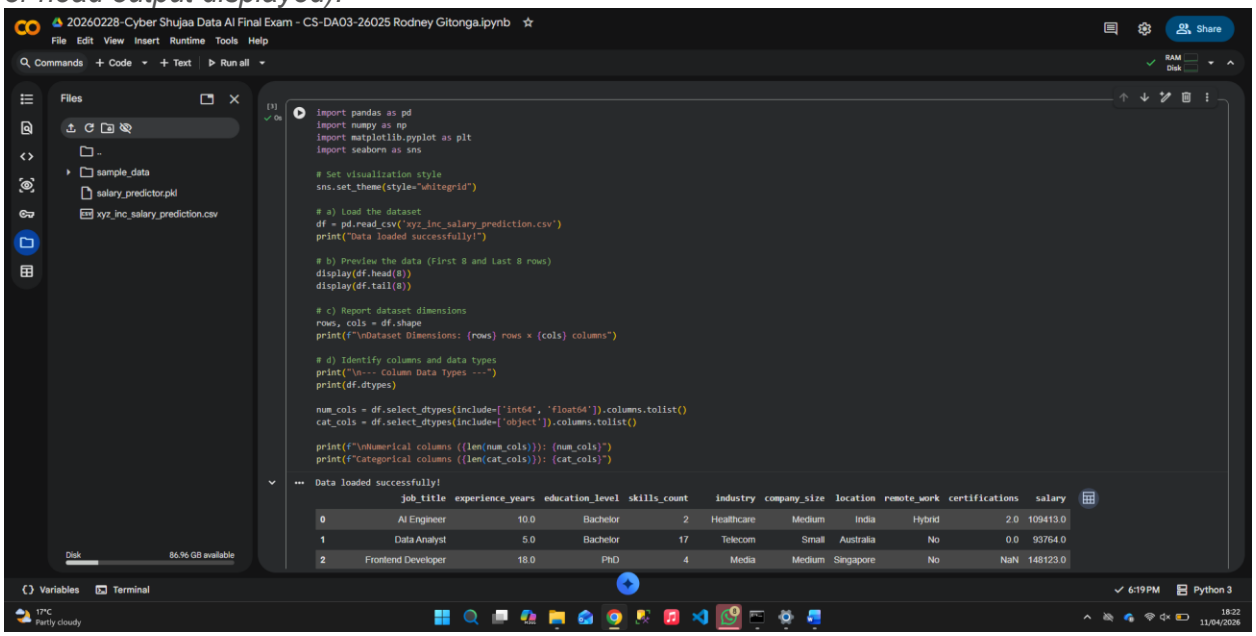
PART 1: Data Engineering & Feature Construction (30 Marks)

1.1 Data Loading and Initial Assessment (10 Marks)

a) Load the dataset. (1 Mark)

Load the xyz_inc_salary_prediction.csv file into your Google Colab notebook using Pandas. Show successful loading with a confirmation output (e.g., print the shape or display .head()).

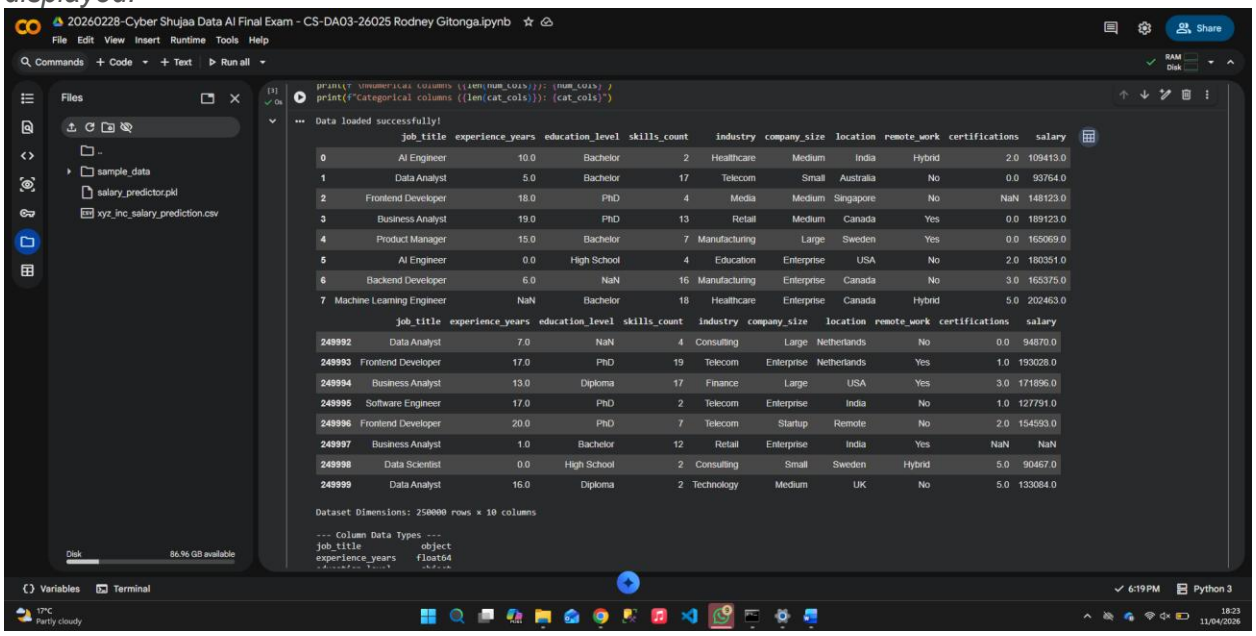
> *Marks: 1 mark for loading the dataset and showing proof of successful loading (e.g., shape or head output displayed).*



b) Preview the data. (1 Mark)

Display the first 8 rows and the last 8 rows of the dataset.

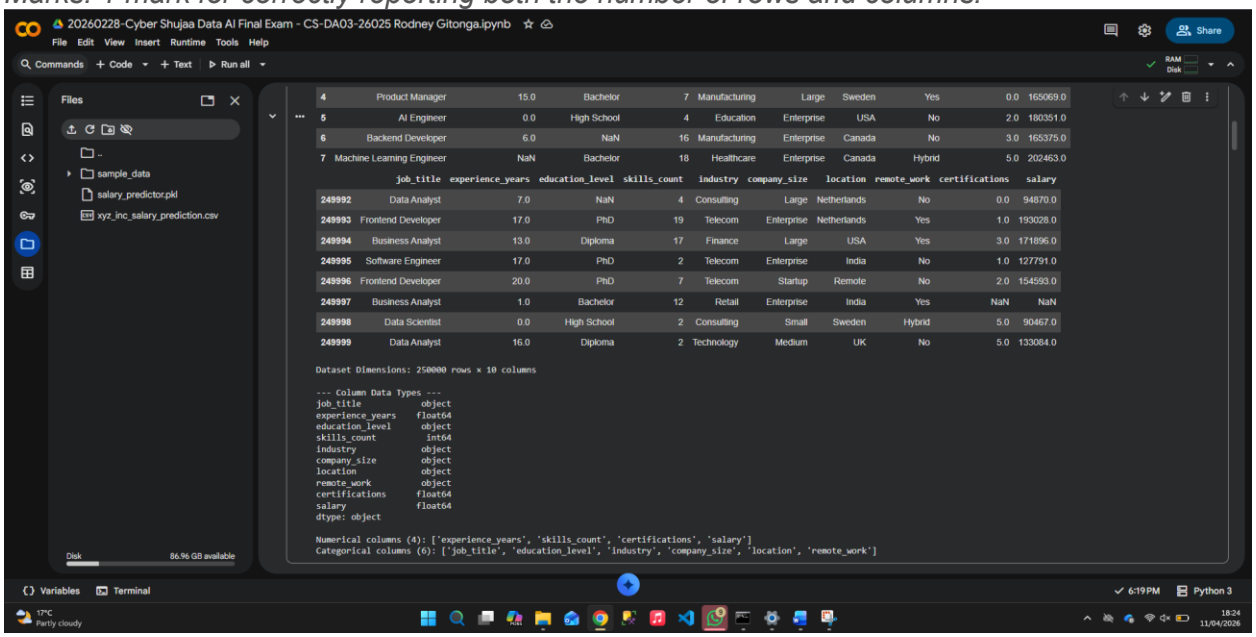
Marks: 0.5 marks for first 8 rows correctly displayed; 0.5 marks for last 8 rows correctly displayed.



c) Report the dataset dimensions. (1 Mark)

State the total number of rows and columns in the format: rows × columns.

Marks: 1 mark for correctly reporting both the number of rows and columns.

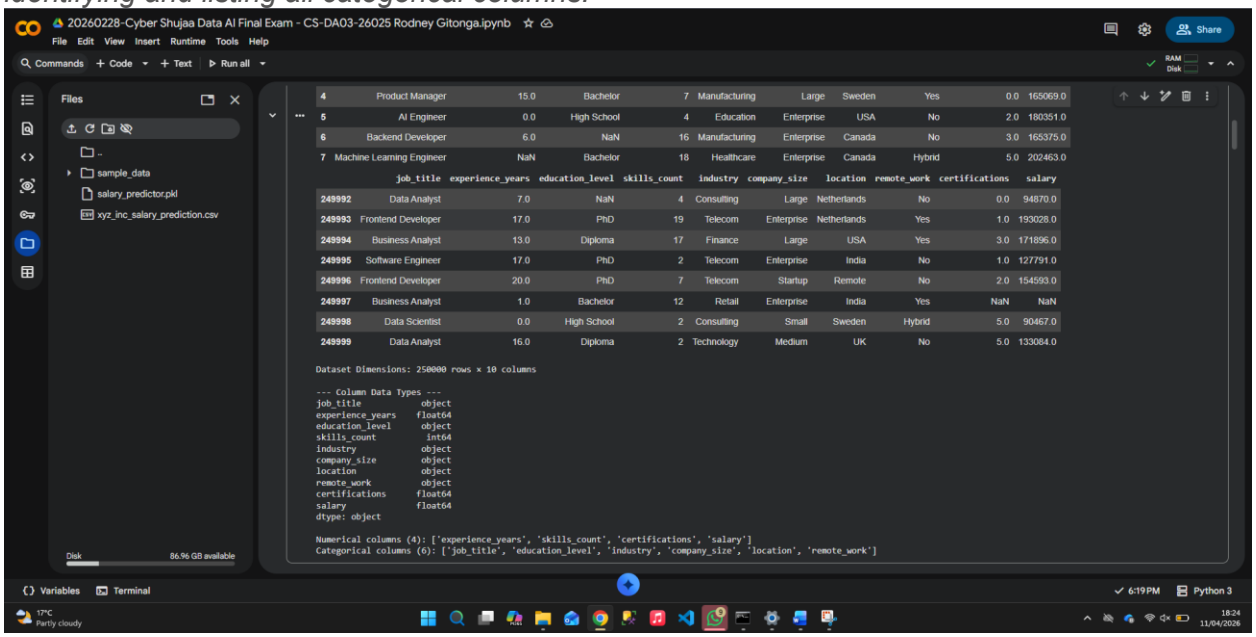


d) Identify columns and data types. (2 Marks)

List all columns and their data types. Then answer:

- How many numerical columns does the dataset contain? Name them. (1 Mark)
- How many categorical columns does the dataset contain? Name them. (1 Mark)

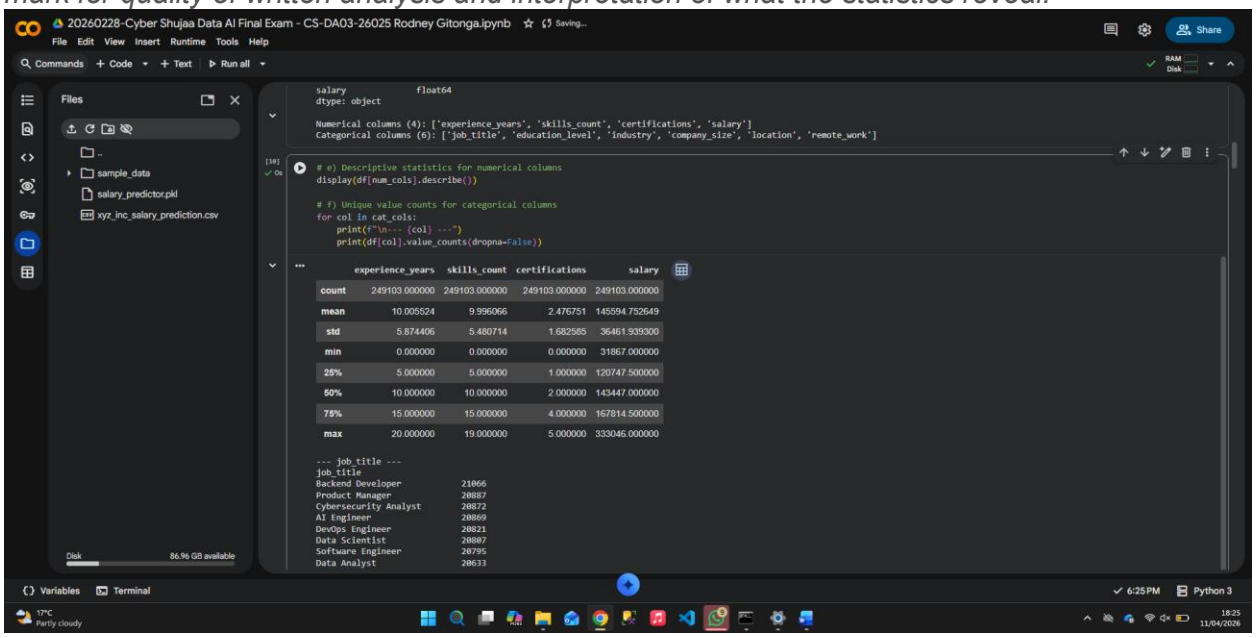
Marks: 1 mark for correctly identifying and listing all numerical columns; 1 mark for correctly identifying and listing all categorical columns.



e) Descriptive statistics for numerical columns. (3 Marks)

Compute statistical summaries (mean, median, min, max, standard deviation, and quartiles) for all numerical columns: experience_years, skills_count, certifications, and salary. Analyse and explain your findings in a Markdown cell — what do these statistics reveal about the workforce and compensation patterns in the dataset?

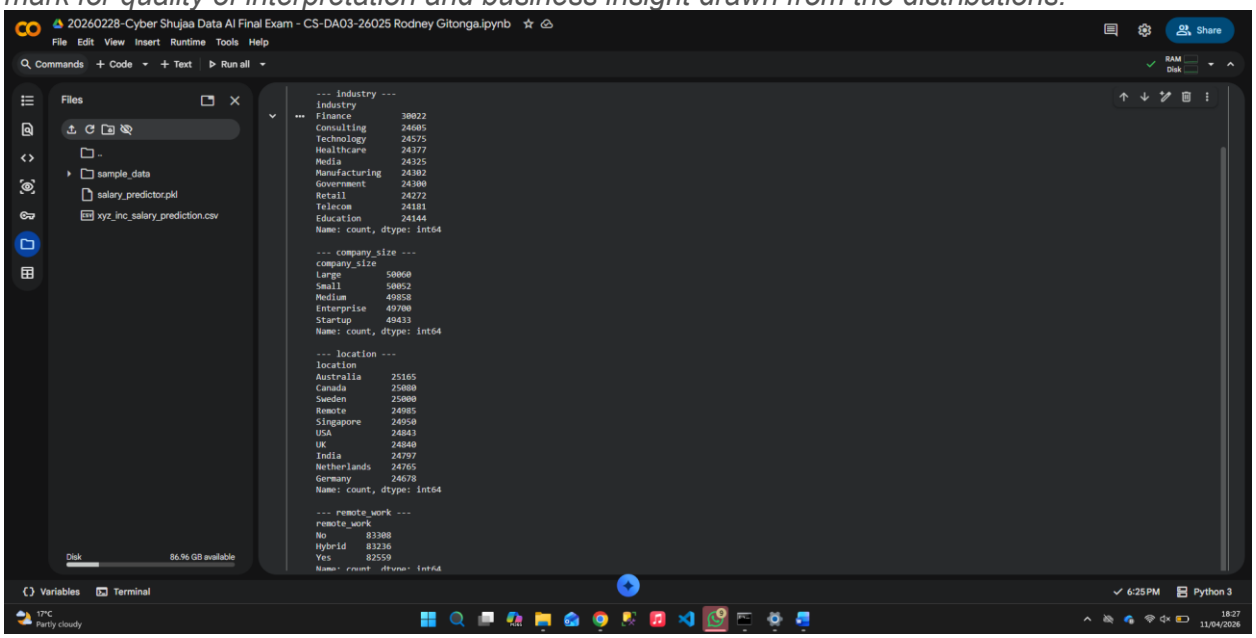
Marks: 2 marks for correctly computing descriptive statistics for all four numerical columns; 1 mark for quality of written analysis and interpretation of what the statistics reveal.



f) Unique value counts for categorical columns. (3 Marks)

Display the unique value counts for all categorical columns: job_title, education_level, industry, company_size, location, and remote_work. Analyse and explain what insights these distributions reveal about the nature of the workforce represented in this dataset.

> *Marks: 2 marks for correctly displaying unique value counts for all categorical columns; 1 mark for quality of interpretation and business insight drawn from the distributions.*



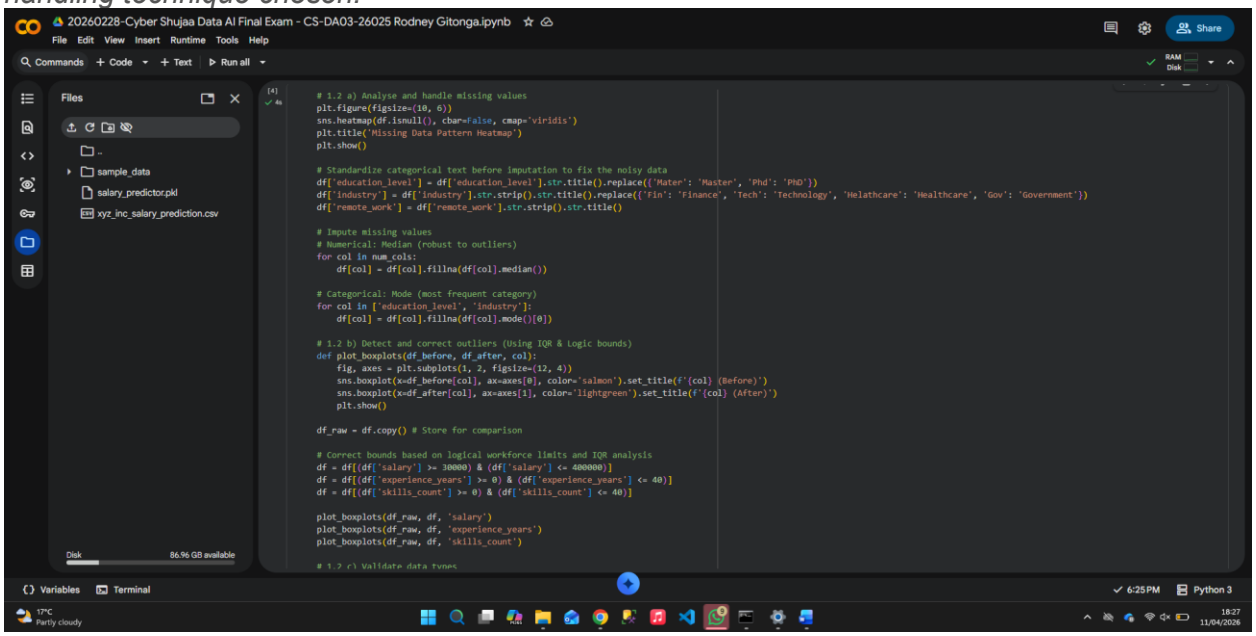
1.2 Data Cleaning (10 Marks)

a) Analyse and handle missing values. (3 Marks)

Check for missing values across all columns. Visualise the missing data pattern (e.g., using a heatmap). Apply appropriate handling techniques for any missing values found and justify your chosen approach. If no missing values exist, document this finding and explain what you would have done had they been present.

> *Marks: 1 mark for visualisation of missing value patterns; 1 mark for correct handling (or documented explanation if none are found); 1 mark for justification of the imputation or*

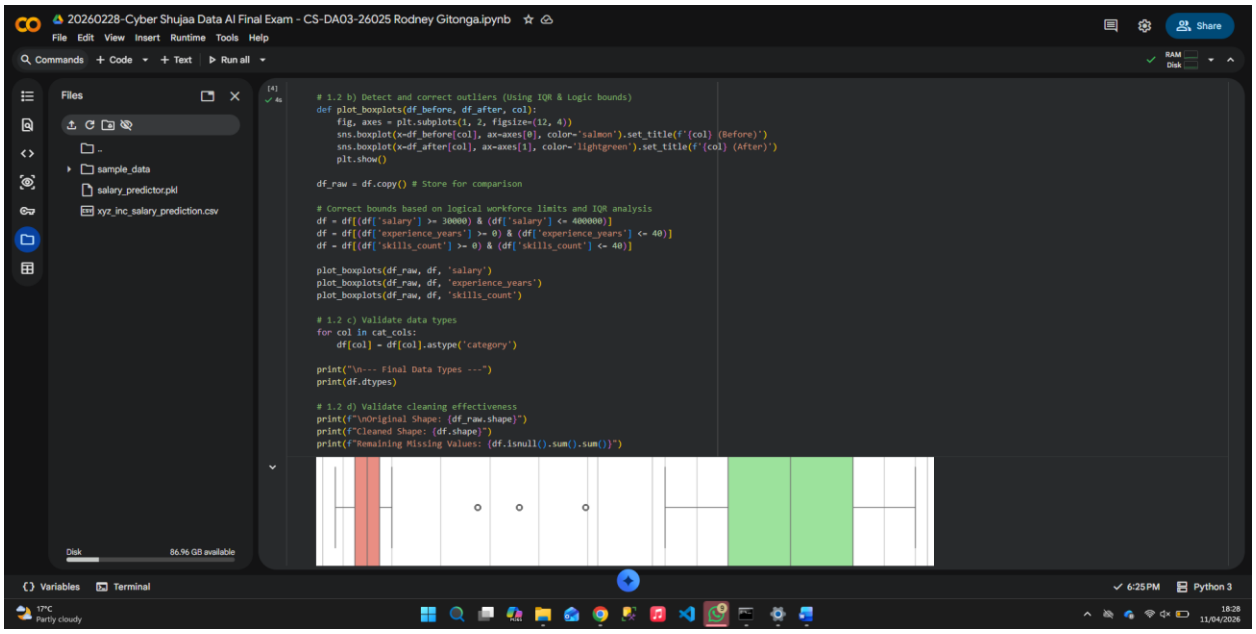
handling technique chosen.



b) Detect and correct outliers. (3 Marks)

Identify outliers in at least three numerical columns (e.g., salary, experience_years, skills_count). Produce before-and-after boxplots to visualise the impact of your correction. Justify the method you used (e.g., IQR capping, z-score removal, or log transformation).

Marks: 1 mark for correct detection of outliers across at least three columns; 1 mark for before-and-after boxplots clearly showing the effect of correction; 1 mark for justification of the chosen outlier treatment method.

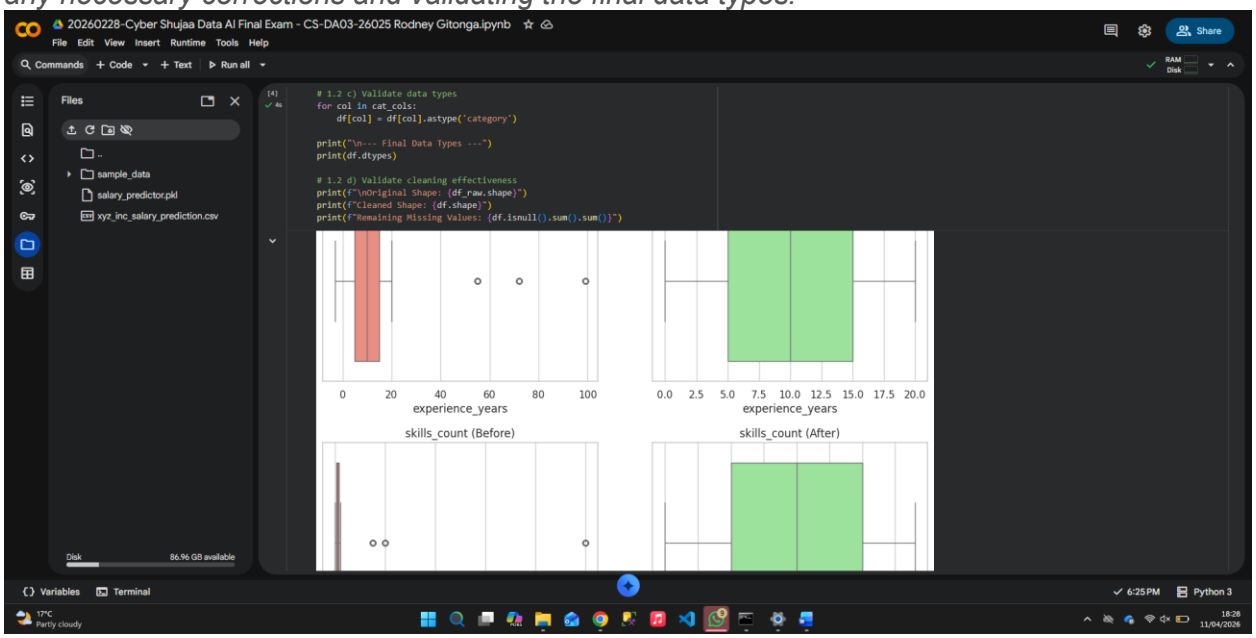


c) Validate data types. (2 Marks)

Confirm that all columns are stored in their correct data types. Ensure numerical columns (experience_years, skills_count, certifications, salary) are integers or floats,

and categorical columns are stored as string or category types. Show the data types before and after any corrections.

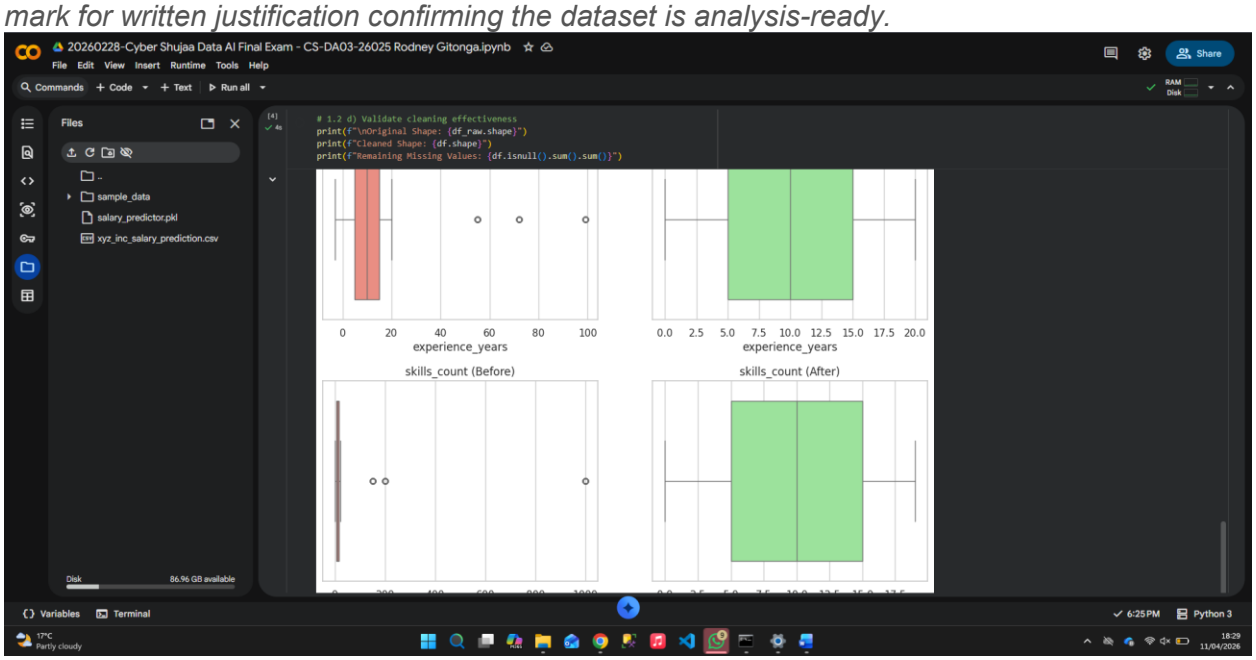
> *Marks: 1 mark for correctly verifying and displaying all column data types; 1 mark for making any necessary corrections and validating the final data types.*



d) Validate overall data cleaning effectiveness. (2 Marks)

Confirm your cleaning is complete using a before-and-after summary comparison (e.g., shape, missing value counts, or descriptive statistics). Write a brief justification explaining that the dataset is now ready for analysis and modelling.

> *Marks: 1 mark for a clear before-and-after comparison showing the effect of cleaning; 1 mark for written justification confirming the dataset is analysis-ready.*



1.3 Feature Engineering (10 Marks)

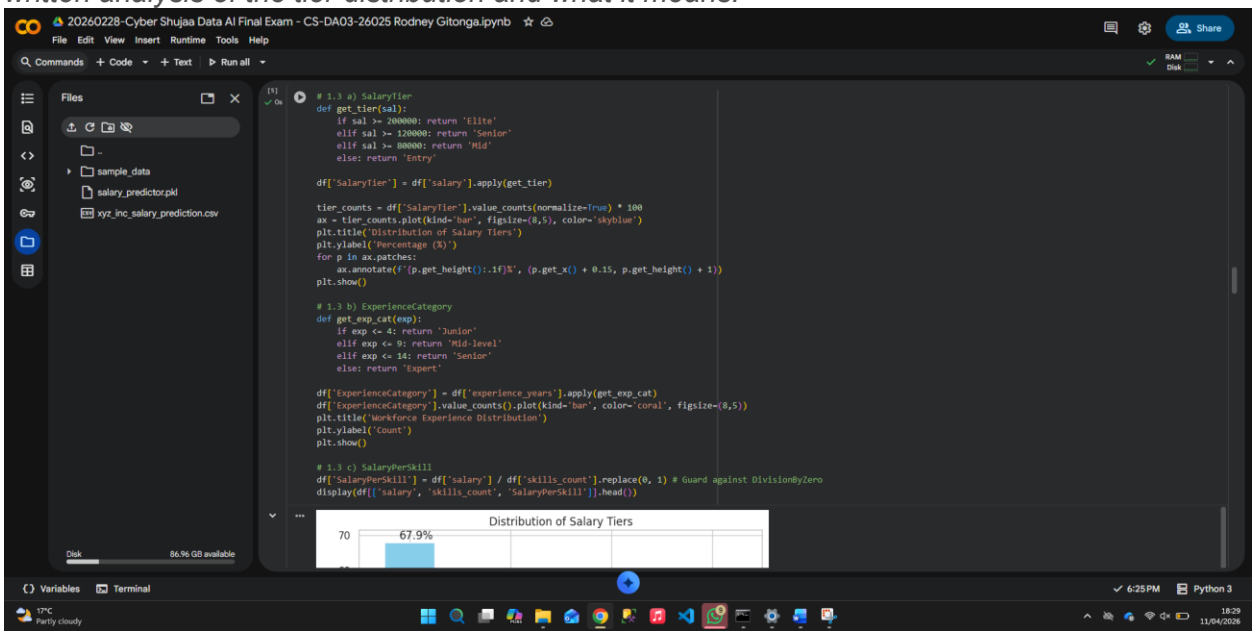
a) SalaryTier- Derived categorical feature. (5 Marks)

Create a new column called SalaryTier by segmenting the salary column using the following thresholds, which are calibrated to this dataset's salary range ($31,867 – $333,046):

Tier	Salary Range (USD)	Approximate Percentile
Elite	≥ $200,000	Top ~10%
Senior	$120,000 – $199,999	50th – 90th
Mid	$80,000 – $119,999	20th – 50th
Entry	< $80,000	Bottom ~20%

After creating the column, produce a distribution plot (bar chart or countplot) showing the count of employees in each SalaryTier. Include percentage labels on the bars. Write a brief analysis of the distribution — are the tiers balanced? What does this reveal about the salary structure?

> *Marks: 2 marks for correctly computing and assigning SalaryTier using the thresholds above; 2 marks for a correctly labelled distribution plot with percentage labels; 1 mark for written analysis of the tier distribution and what it means.*



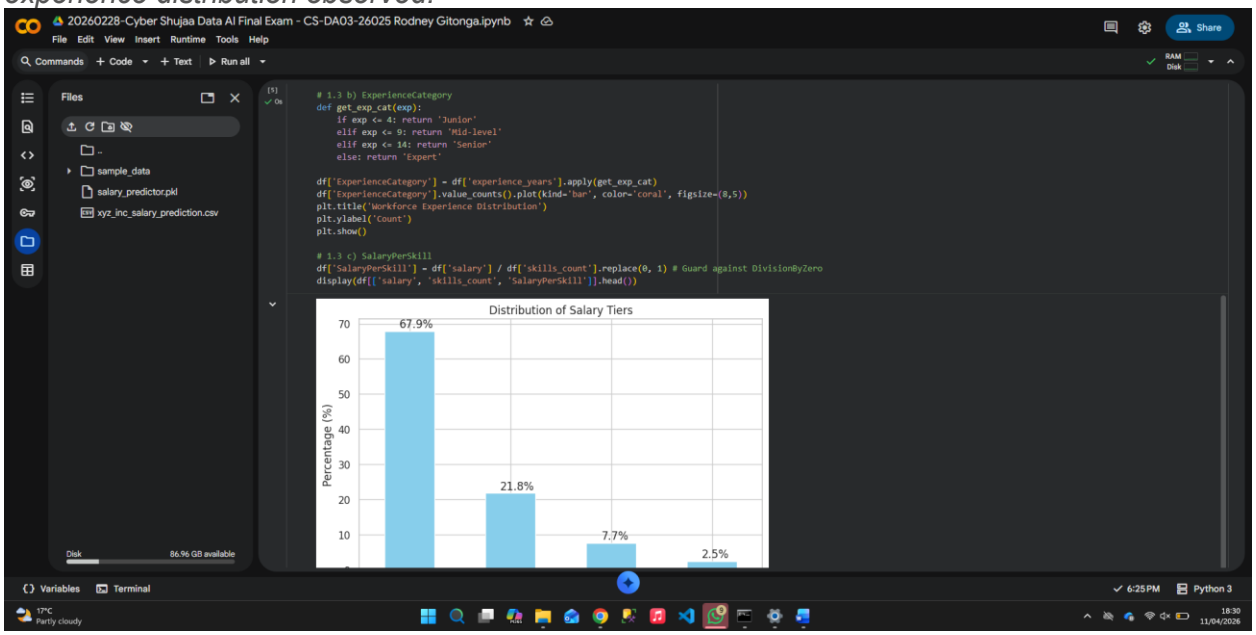
b) ExperienceCategory — Derived categorical feature. (3 Marks)

Create a new column called ExperienceCategory by grouping experience_years into the following bands:

- Junior: 0–4 years
- Mid-level: 5–9 years
- Senior: 10–14 years
- Expert: 15–20 years

Show the value counts for each band and produce a simple bar chart. Explain the logic behind this grouping and what it tells you about the experience distribution across the dataset.

> *Marks: 2 marks for correctly creating ExperienceCategory with the four bands above and displaying value counts; 1 mark for written explanation of the grouping logic and the experience distribution observed.*



c) SalaryPerSkill — Derived numerical feature. (2 Marks)

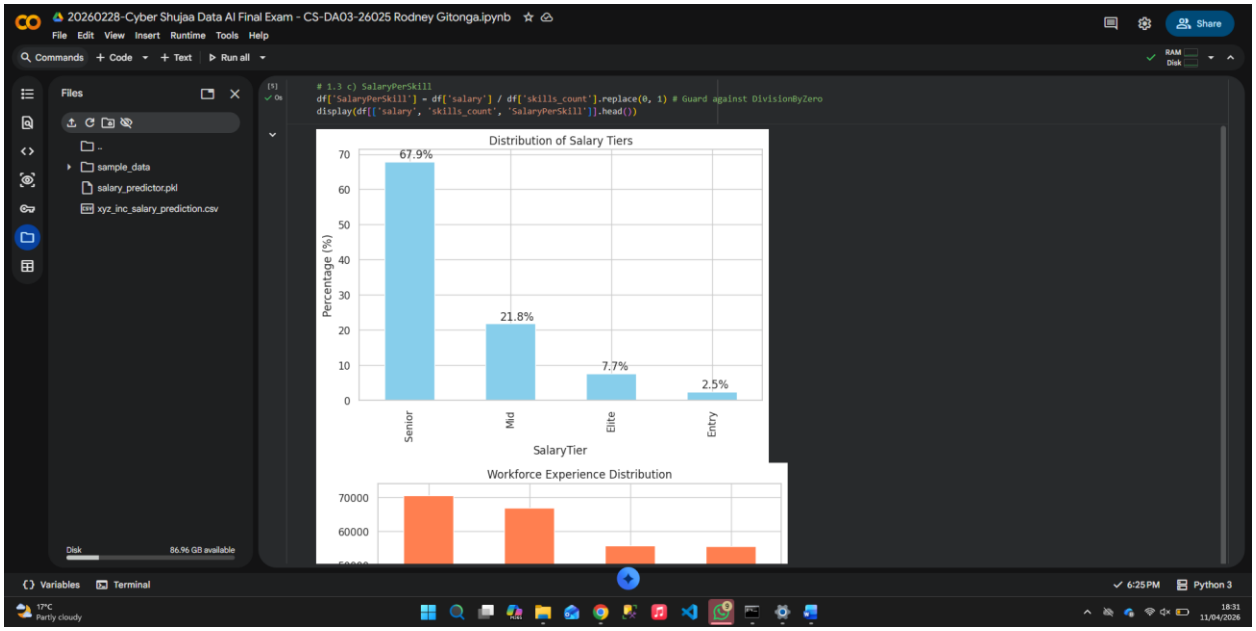
Create a new column called SalaryPerSkill calculated as:

SalaryPerSkill = salary / skills_count

Handle any edge cases appropriately (e.g., if skills_count could be zero, add a guard). Explain what this metric represents and how XYZ Inc. could use it to advise job seekers on the value of acquiring additional skills.

> *Marks: 1 mark for correct calculation of SalaryPerSkill with appropriate edge-case handling; 1 mark for a clear written explanation of what this metric represents and its practical*

business use.



PART 2: Exploratory Data Analysis (20 Marks)

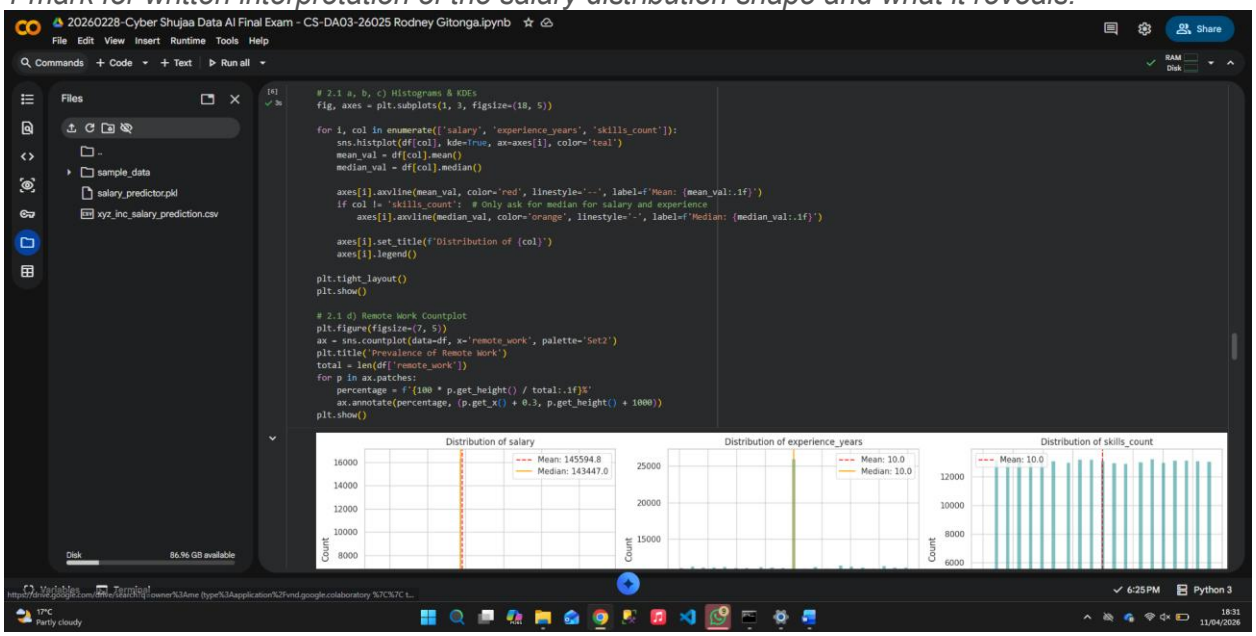
2.1 Univariate Analysis (8 Marks)

Using Seaborn or Matplotlib, create histogram and KDE (density) plots for the four variables below. Each plot must include statistical annotations showing the mean, median, and standard deviation. Follow each plot with a written interpretation in a Markdown cell.

a) Salary (salary) (2 Marks)

Plot a histogram with KDE overlay for the salary column. Annotate the mean and median. What does the shape of the distribution (e.g., skewed, normal, bimodal) tell you about pay distribution across the workforce in this dataset?

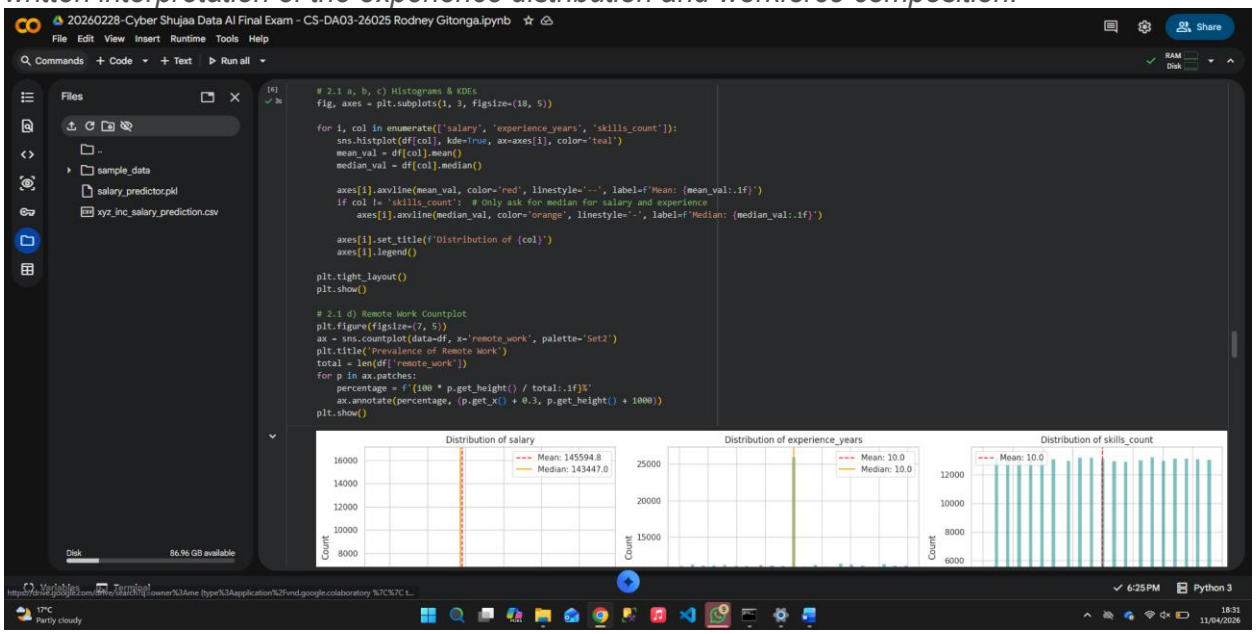
> *Marks: 1 mark for a correctly labelled histogram/KDE plot with mean and median annotated; 1 mark for written interpretation of the salary distribution shape and what it reveals.*



b) Experience Years (experience_years) (2 Marks)

Plot a histogram with KDE overlay for experience_years. Annotate the mean and median. Interpret the distribution — is the workforce predominantly junior, mid-level, or senior?

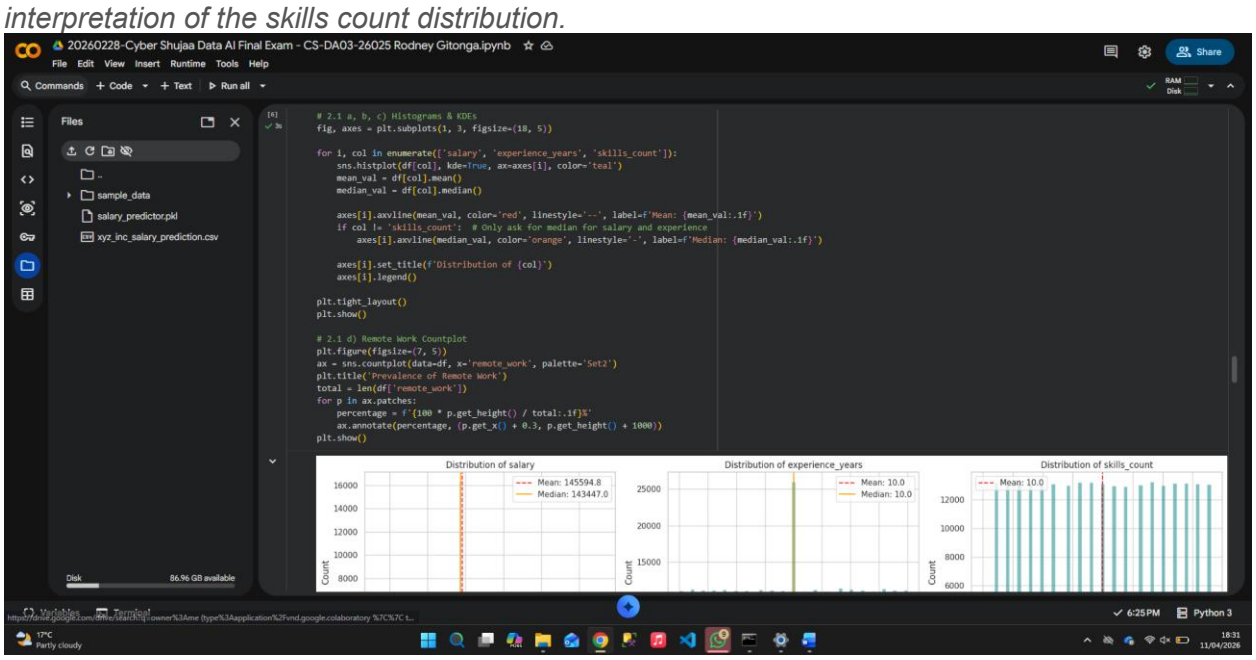
Marks: 1 mark for a correctly labelled histogram/KDE plot with annotations; 1 mark for written interpretation of the experience distribution and workforce composition.



c) Skills Count (skills_count) (2 Marks)

Plot a histogram with KDE overlay for skills_count. Annotate the mean. What does this distribution tell you about the breadth of technical skills across employees in the dataset?

Marks: 1 mark for a correctly labelled histogram/KDE plot with annotation; 1 mark for written interpretation of the skills count distribution.

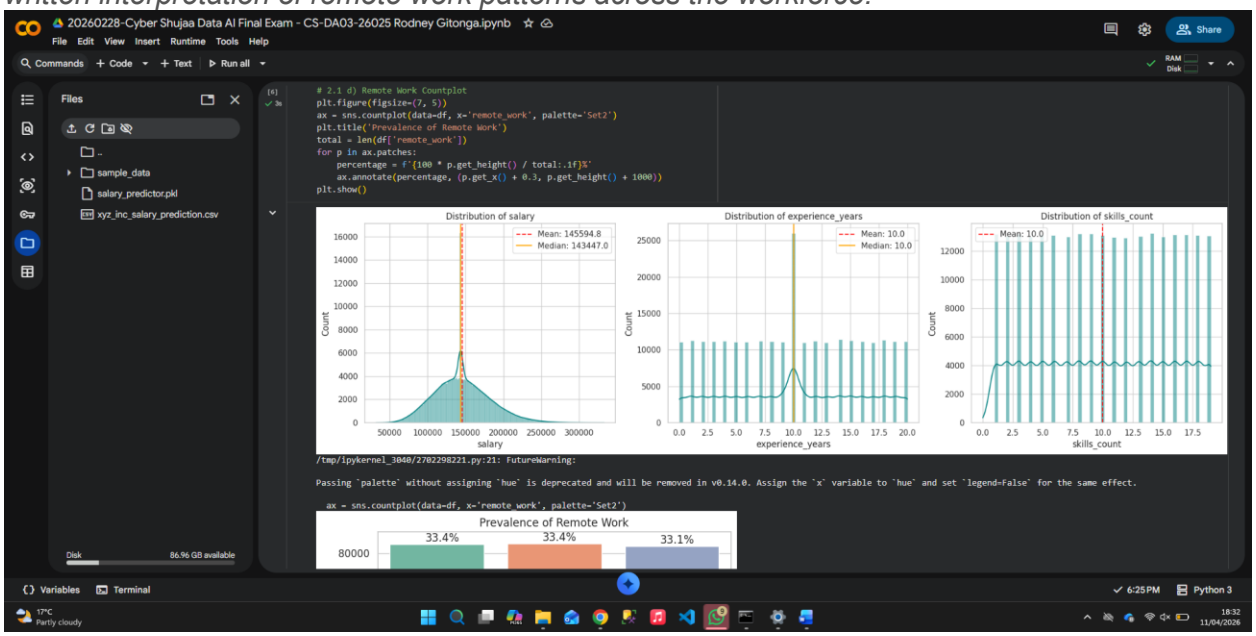


d) Remote Work Arrangement (remote_work) (2 Marks)

Since remote_work is a categorical column (Yes / No / Hybrid), produce a countplot showing the frequency of each work arrangement. Calculate and annotate the

percentage for each category. What does this tell you about the prevalence of remote and hybrid working in the dataset?

> *Marks: 1 mark for a correctly labelled countplot with percentage annotations; 1 mark for written interpretation of remote work patterns across the workforce.*



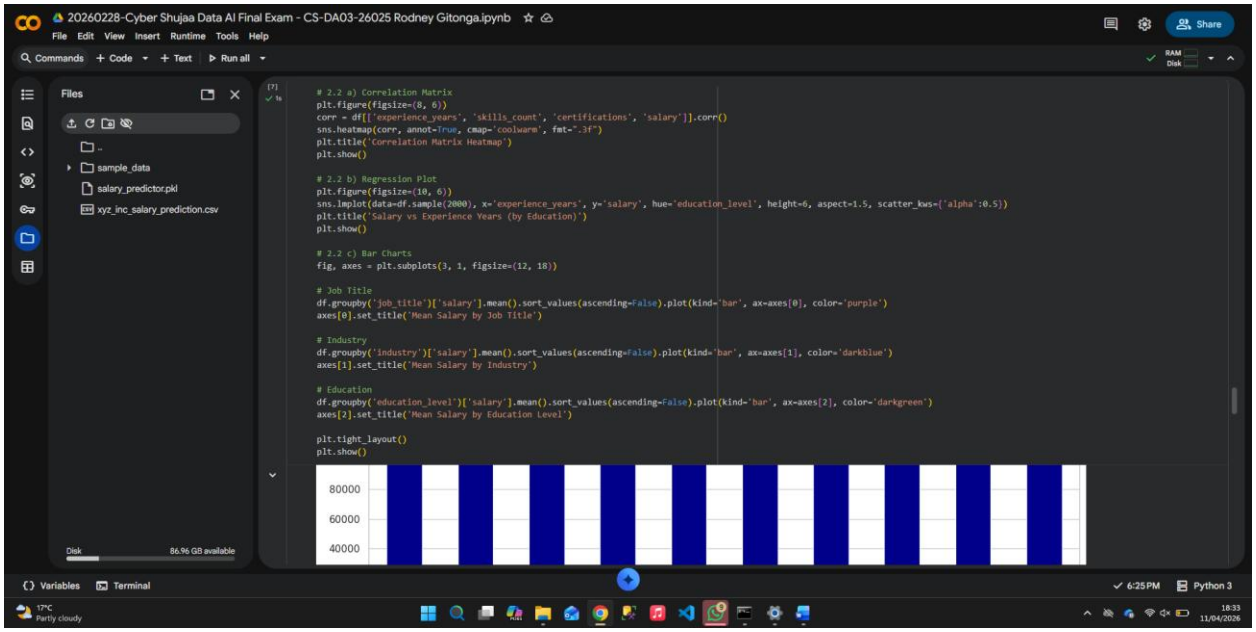
2.2 Bivariate and Multivariate Analysis (12 Marks)

a) Correlation matrix heatmap. (4 Marks)

Compute and plot a correlation matrix heatmap for all numerical columns: experience_years, skills_count, certifications, and salary. Annotate the heatmap with correlation coefficients. Then analyse your findings: Which pairs of variables are most strongly correlated with salary? Are any correlations surprising or counterintuitive? What do the correlations suggest about what drives compensation in this dataset?

> *Marks: 2 marks for a correctly computed and clearly annotated correlation heatmap; 2 marks for written analysis that identifies the strongest correlations and explains their meaning in a*

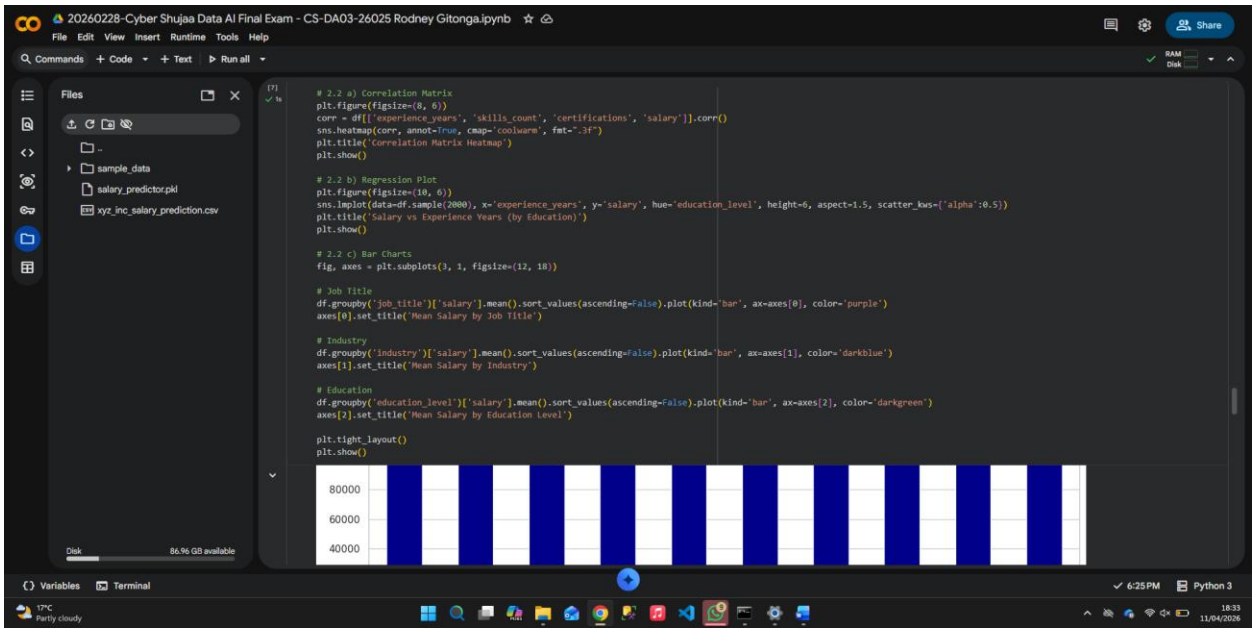
business context.



b) Regression plot: Salary vs Experience Years. (3 Marks)

Create a regression scatter plot (with trend line) showing the relationship between salary and experience_years. Colour-code the points by education_level to add a multivariate dimension. Explain: What does the slope of the trend line tell you? Does education level appear to modify the salary-experience relationship?

Marks: 2 marks for a correctly produced regression plot with colour-coding by education_level; 1 mark for written explanation of the relationship observed and how education interacts with it.



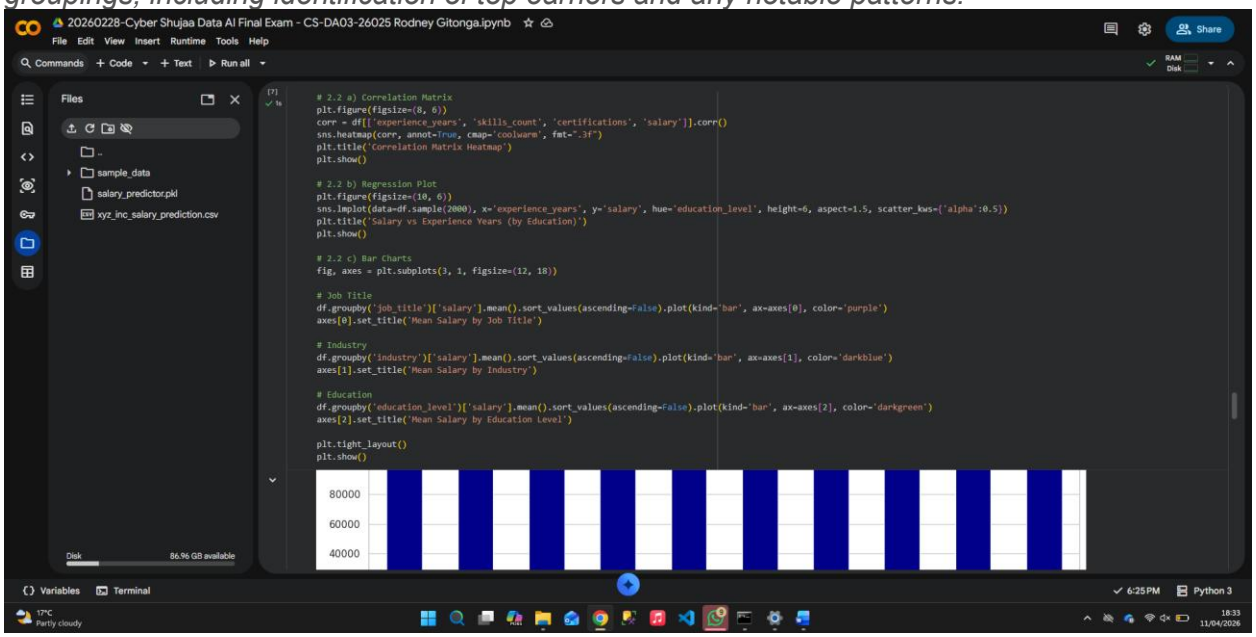
c) Bar charts: Salary patterns across job categories and industries. (5 Marks)

Using groupby, produce bar charts showing the mean salary for each of the following groupings. Include at least three separate charts:

- Mean salary by job_title (sorted descending)
- Mean salary by industry
- Mean salary by education_level

For each chart, explain your findings: Which job titles command the highest salaries? Which industries pay the most? How does education level affect compensation?

> *Marks: 3 marks for correctly produced and labelled bar charts — 1 mark each for job_title, industry, and education_level; 2 marks for written explanation of salary trends across these groupings, including identification of top earners and any notable patterns.*



PART 3: Model Development & Deployment (50 Marks)

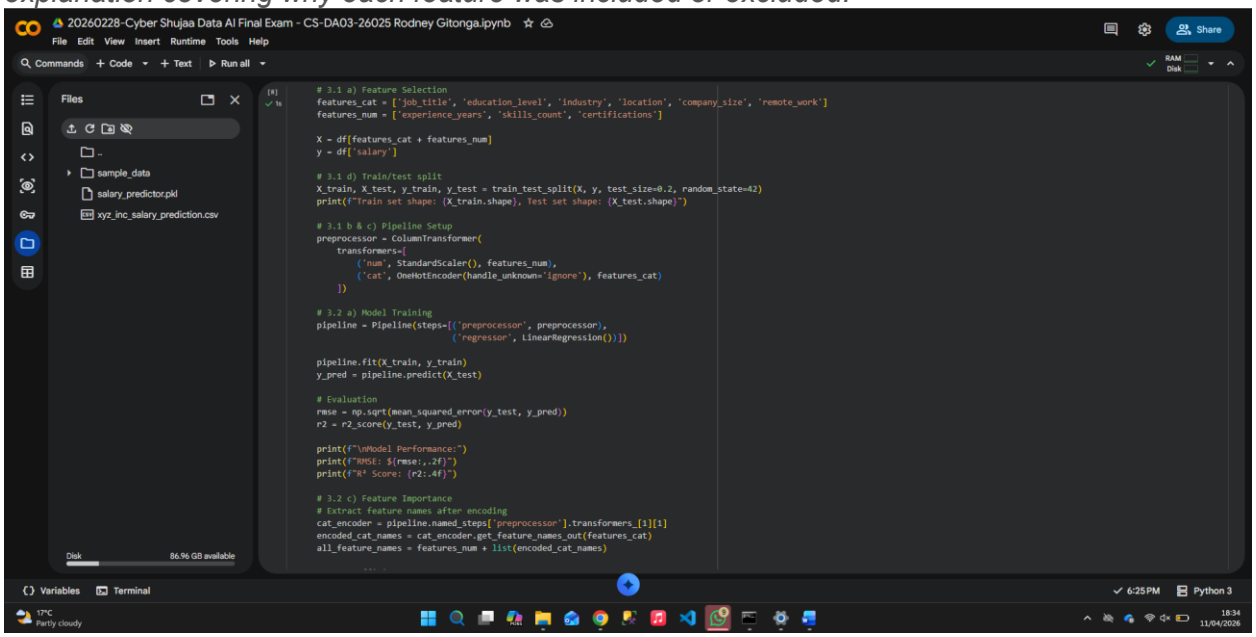
3.1 Data Preprocessing (15 Marks)

Using Scikit-learn, implement a full preprocessing pipeline before model training.

a) Feature selection and justification. (5 Marks)

Identify and justify the features you will use for model building. Select at least three categorical variables and at least three numerical variables from the dataset. For each selected feature, explain why it is likely to be a meaningful predictor of salary. Also state any columns you are excluding and why (e.g., to avoid data leakage or because they are redundant).

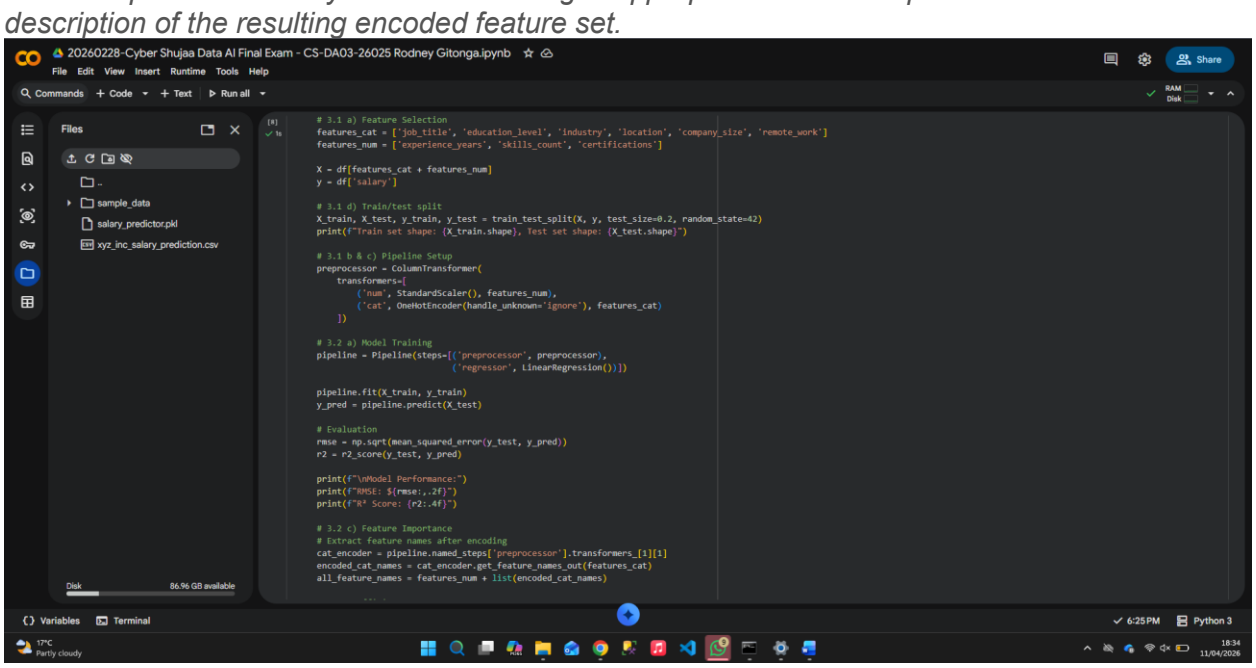
Marks: 3 marks for a correct and well-reasoned feature selection 2 marks for written explanation covering why each feature was included or excluded.



b) One-hot encoding for categorical variables. (4 Marks)

Apply One-Hot Encoding to your chosen categorical variables using Scikit-learn's OneHotEncoder (within a Pipeline or ColumnTransformer). Explain: Why is one-hot encoding appropriate for these variables rather than label encoding? Show a sample of the encoded output — how many new columns were created? What do they represent?

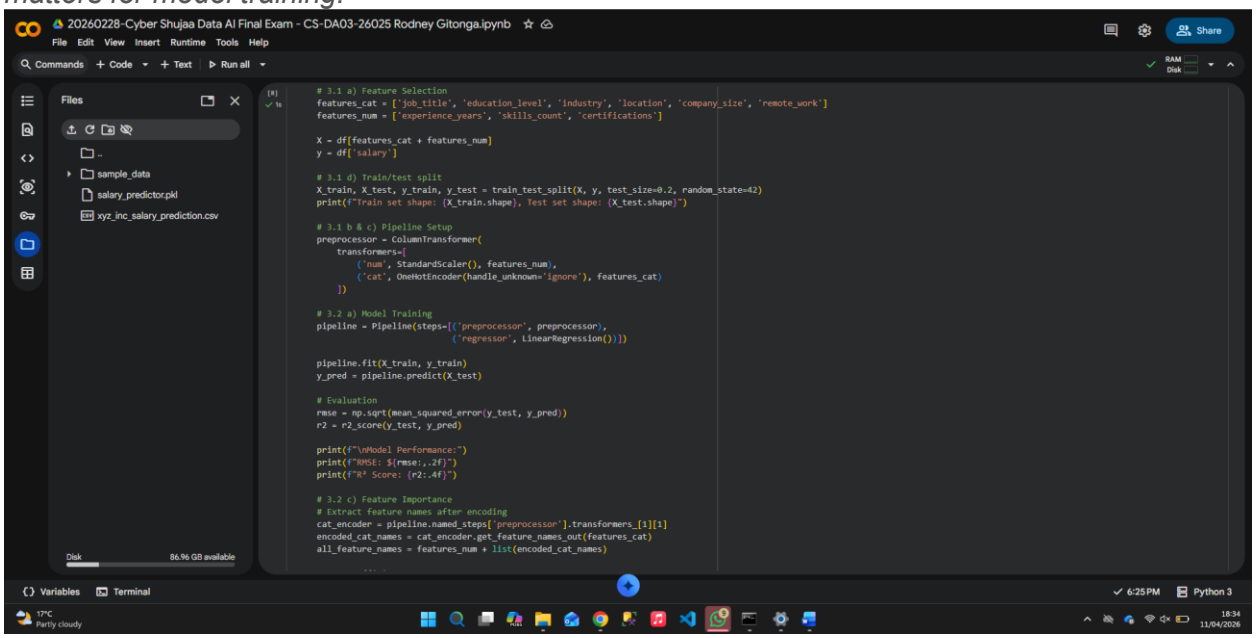
Marks: 2 marks for correct implementation of one-hot encoding within a pipeline; 2 marks for written explanation of why one-hot encoding is appropriate for these specific variables and a description of the resulting encoded feature set.



c) Standard scaling for numerical variables. (4 Marks)

Apply StandardScaler to your chosen numerical variables (e.g., experience_years, skills_count, certifications). Show before-and-after summary statistics (mean and standard deviation). Explain: What does StandardScaler do to the values mathematically? Why is feature scaling important when training a predictive model?

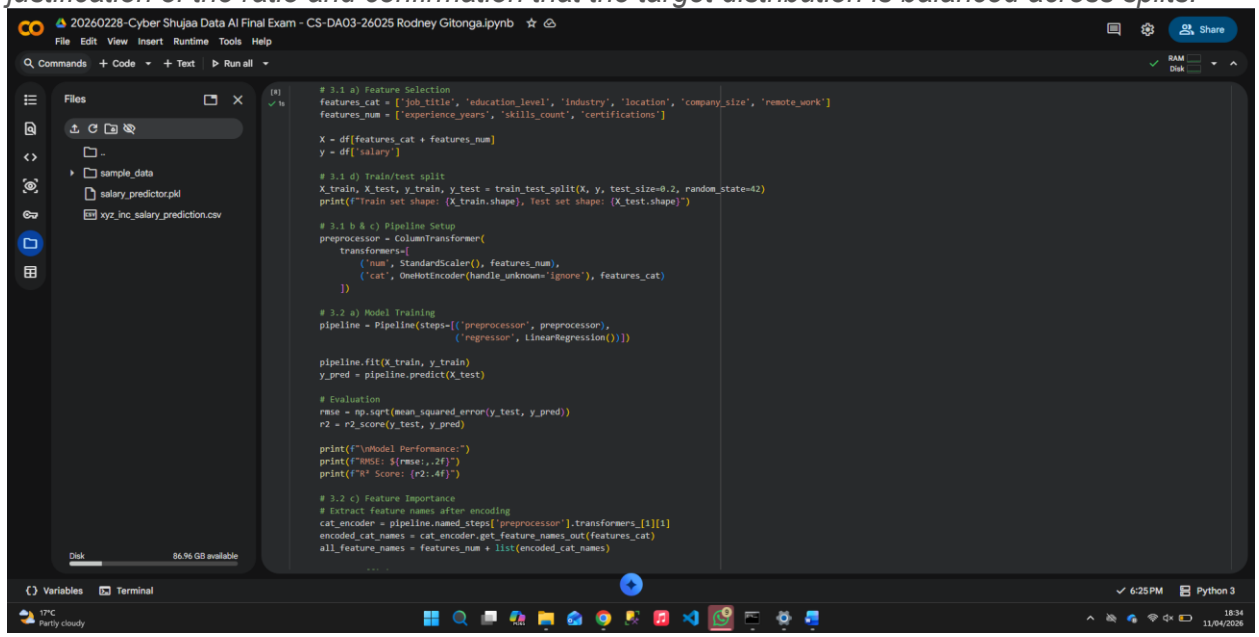
> *Marks: 2 marks for correct implementation of StandardScaler and showing before-and-after statistics; 2 marks for written explanation of what scaling does mathematically and why it matters for model training.*



d) Train/test split. (2 Marks)

Split the dataset into training and test sets. Justify your chosen split ratio (e.g., 80/20 or 70/30). Report the shape of each set.

> *Marks: 1 mark for performing the split and reporting the shape of each set; 1 mark for justification of the ratio and confirmation that the target distribution is balanced across splits.*



3.2 Model Development (10 Marks)

Build **a Linear Regression model** to predict the continuous salary column.

a) Train your chosen model. (4 Marks)

Train a LinearRegression model to predict salary. Evaluate performance on both training and test sets using RMSE and R^2. Interpret your results: Is the model performing well? Is there evidence of overfitting or underfitting? What does the R^2 value mean in plain language for a business stakeholder?

> *Marks: 4 marks for correctly training the model: 2 marks for correct implementation and fit; 2 marks for computing and displaying all required evaluation metrics.*

b) Interpret your model results. (4 Marks)

Write a detailed interpretation of your model's performance. Address all of the following:

- Are the results satisfactory? Why or why not?
- Where is the model performing well and where is it struggling? (e.g., which salary tier is hardest to classify, or where are the largest prediction errors?)
- What does the performance mean in plain language for a non-technical business stakeholder at XYZ Inc.?
- What are two specific steps you could take to improve the model's performance?

> *Marks: 2 marks for a thorough, evidence-backed interpretation that references specific metric values; 2 marks for the plain-language explanation and two concrete, realistic improvement suggestions.*

c) Feature importance. (2 Marks)

Identify and visualise the top 5 most important features driving your model's predictions. Use the absolute values of the model's coefficients to rank feature importance. Plot a horizontal bar chart of the top 5 features ranked by importance. Explain: What do these features tell you about what drives salary levels in this dataset?

> *Marks: 1 mark for a correctly produced and labelled feature importance bar chart showing the top 5 features; 1 mark for written explanation of what the top features reveal about salary drivers.*

3.3 Business Insights (10 Marks)

Drawing on your analysis from Parts 1, 2, and 3, identify and document at least five data-driven insights that can help XYZ Inc. improve its salary benchmarking service, career advisory product, or talent acquisition offering.

Structure each insight using the following format:

- Finding: What did the data show?
- Evidence: Which specific plot, statistic, or model output supports this finding?
- Recommendation: What should XYZ Inc. do in response to this finding?

- Expected Impact: What measurable business benefit could this deliver?

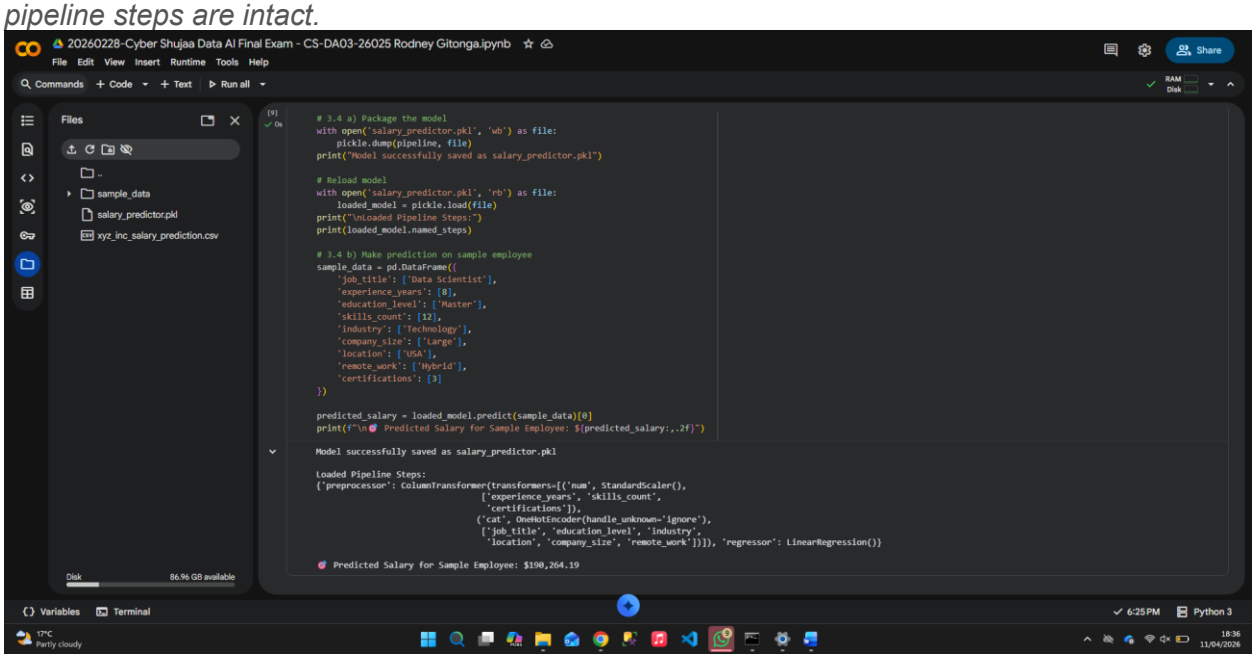
> *Marks: 2 marks per insight: 1 mark for a valid, data-backed finding with cited evidence; 1 mark for a practical, specific recommendation with a clearly stated expected business impact. (5 insights × 2 marks = 10 marks total.)*

3.4 Model Packaging & Deployment (15 Marks)

a) Package your trained model. (3 Marks)

Save your trained model together with its complete preprocessing pipeline (ColumnTransformer / Pipeline + trained model) into a single .pkl file using Python's pickle library. Show the code used to save the file. Then reload the .pkl file in a new cell and confirm it loads correctly by printing the pipeline steps.

> *Marks: 1 mark for correctly packaging the full preprocessing pipeline with the trained model; 1 mark for saving it to a .pkl file; 1 mark for reloading the file in a new cell and confirming the pipeline steps are intact.*



b) Make a prediction on a sample employee record. (4 Marks)

Use your loaded model to predict the salary for the following sample employee:

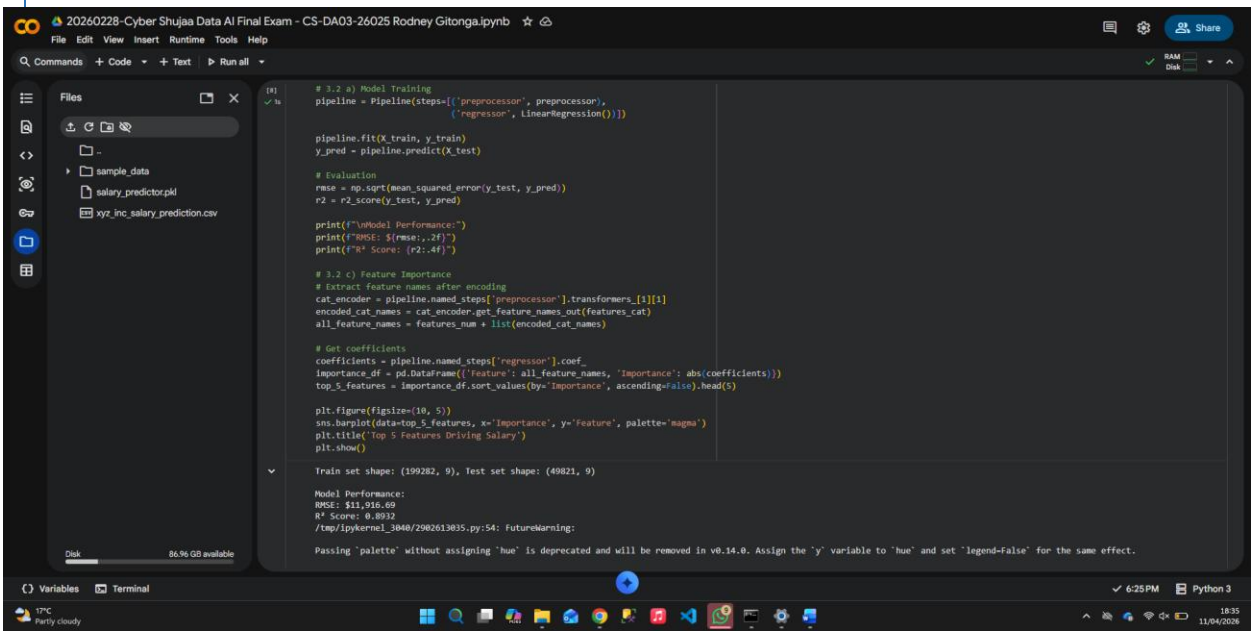
Feature	Sample Value	Notes
job_title	**Data Scientist**	*One of the 12 valid job titles*
experience_years	**8**	*Mid-level experience*

education_level	**Master**	*Postgraduate qualification*
skills_count	**12**	*12 technical skills*
industry	**Technology**	*Tech sector*
company_size	**Large**	*Large employer*
location	**USA**	*United States*
remote_work	**Hybrid**	*Partial remote*
certifications	**3**	*Three certifications held*

You must:

- State the predicted salary.
- Display the exact predicted salary value.
- Include a full-screen screenshot of the prediction output in your notebook.
- Explain in plain language what the prediction means — is this a competitive outcome for a Data Scientist in the USA with this profile? Does it align with what you would expect based on your EDA findings?

> *Marks: 1 mark for a correct and working prediction from the loaded model; 1 mark for displaying the exact predicted salary value; 1 mark for a screenshot of the prediction output; 1 mark for a clear, well-reasoned plain-language explanation of the result.*



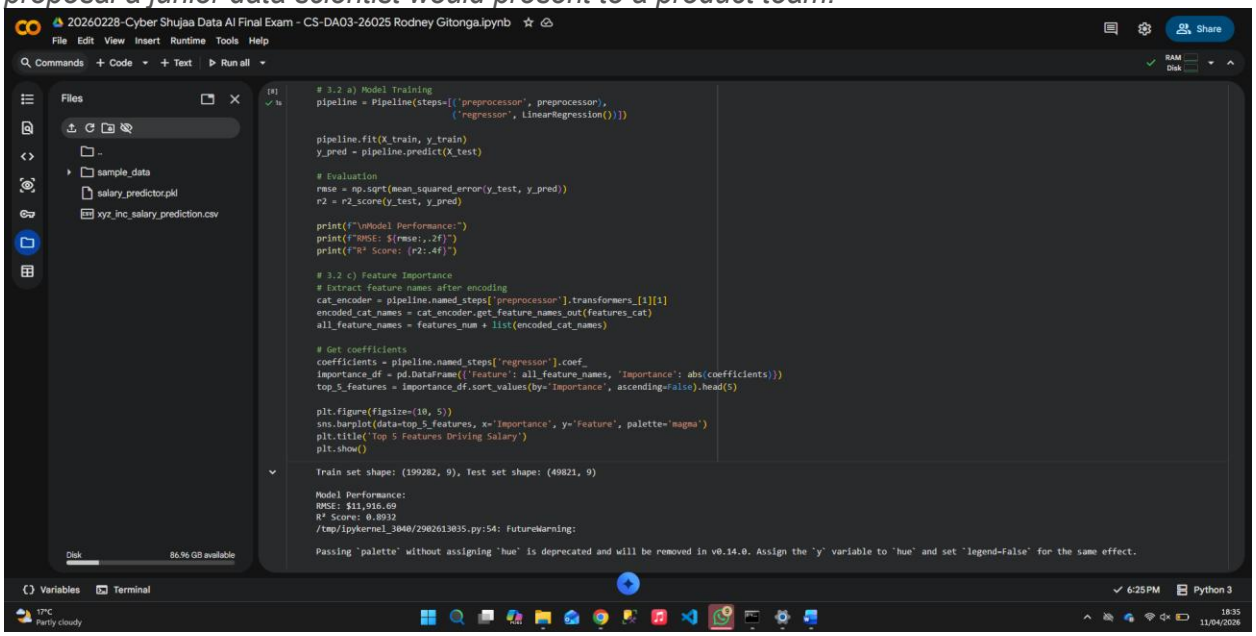
c) Deployment plan. (8 Marks)

Write a detailed, realistic plan for how XYZ Inc. could deploy and consume your packaged model in a production environment to power its salary benchmarking product. Your plan must clearly address all six of the following areas:

Area	What to address
API deployment	How would you expose the model as a REST API?
Input validation	How would you validate and preprocess incoming user data at inference time before passing it to the model?
Model monitoring	How would you detect model drift or degradation in prediction quality over time?
Retraining strategy	How and when would you retrain the model as new salary data becomes available?
Security & access control	How would you protect the API from unauthorised access and ensure user data privacy?
Platform integration	How would the model integrate with XYZ Inc.'s existing job portal or HR analytics dashboard?

> *Marks: 5 marks for a logical, realistic plan that addresses all six areas with sufficient technical detail (deduct 1 mark per area that is missing or too vague); 3 marks for overall clarity, structure, and quality of written explanation — a strong answer reads like a technical*

proposal a junior data scientist would present to a product team.



SUBMISSION INSTRUCTIONS

Upload all required files to the LMS before leaving the exam. Incomplete submissions will be penalised.

#	File / Item	Where to Submit
1	.pdf report generated from this document — your primary grading document	LMS upload
2	.pkl file of your best packaged model (with preprocessing pipeline)	LMS upload
3	.ipynb Notebook downloaded from Google Colab with complete code and Markdown explanations	LMS upload

Mark Distribution Summary

Section / Part	Marks	Score
Section A: Multiple Choice Questions (on LMS)	30	

Part 1: Data Engineering & Feature Construction	30	
1.1 Data Loading and Initial Assessment	10	
1.2 Data Cleaning	10	
1.3 Feature Engineering	10	
Part 2: Exploratory Data Analysis	20	
2.1 Univariate Analysis	8	
2.2 Bivariate & Multivariate Analysis	12	
Part 3: Model Development & Deployment	50	
3.1 Data Preprocessing	15	
3.2 Model Development (one model)	10	
3.3 Business Insights	10	
3.4 Model Packaging & Deployment	15	
TOTAL	**100**	

END OF EXAMINATION

Good luck!